CODEALPHA FULL STACK / WEB DEV INTERNSHIP

NEXUS E-Commerce Store

Modern Glassmorphic Client-Side E-Commerce Web Application

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Internship Organization:
CodeAlpha (Virtual Internship
Program)
Recognized by Ministry of Corporate
Affairs, GoI

Project Period:
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Task 1: Simple E-Commerce Store

Key Highlights: 100% Pure HTML5, CSS3 & Vanilla JavaScript | Zero-Server Dependency | LocalStorage Persistence | Full INR (■) Currency Localization | Mobile-First Responsive
Design

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EXECUTIVE SUMMARY

1. Project Overview & Problem Statement

The Objective:

To build a high-performance, visually stunning e-commerce web application that
delivers a frictionless shopping experience directly in the browser with zero
deployment friction.

The Problem Addressed:

• Most traditional demo projects require complex backend runtime setups (databases, servers, pip/npm installs) that fail during quick viva or classroom demos.
• Conventional basic student web pages lack modern UX expectations such as live instant search, category filtering, persistent carts, and checkout confirmation.
• Mobile responsiveness is often an afterthought, resulting in broken layouts on smartphones.

The Solution — NEXUS STORE:

• Pure Client-Side Architecture: 100% executable by opening index.html in any modern browser.
• Full E-Commerce Lifecycle: Catalog browsing ➔ Real-time filtering ➔ Quick View ➔ Interactive Cart Drawer ➔ Checkout Form ➔ Order History ➔ User Authentication.
• Browser-Native State Persistence: Uses HTML5 Web Storage (localStorage) to
emulate an enterprise database.
• Indian Rupee (■) Locale: Realistic INR pricing and formatting.

Deployment Readiness:

Directly hosted and runnable via GitHub Pages with zero cloud hosting costs.

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SYSTEM DESIGN

2. Architecture & Data Flow

1. Presentation Layer (UI) 2. Business Logic Layer 3. Persistence Layer (DB)

• Semantic HTML5 Elements
• Dark Glassmorphism CSS3
• Responsive Flexbox & CSS Grid
• Keyframe Transitions & Modals
• Mobile Viewport Optimizations

• Vanilla JavaScript (ES6+)
• DOM Event Listeners & Rendering
• Search & Category Query Engine
• INR (■) Financial Formatting
• Form Validation & Order Engine

• Browser localStorage API
• nexus_products: Catalog
• nexus_cart: Active Items
• nexus_orders: Placed Bills
• nexus_users: Auth Records

End-to-End User Interaction Flow:

1. Page Load ➔ JS checks localStorage version ➔ Seeds default INR tech products ➔ Renders category chips & cards.
2. Search/Filter ➔ Immediate JS query filter on in-memory array ➔ Dynamic DOM replacement without page refresh.
3. Add to Cart ➔ Item serialized to nexus_cart ➔ Slide-out cart drawer smoothly opens ➔ Subtotal recomputed.
4. Checkout ➔ Multi-field shipping form validated ➔ 6-digit Order ID generated ➔ Saved to nexus_orders ➔ Cart flushed.
5. Order History ➔ Logged-in user accesses historical receipts with date, items, total, and live 'Processing' status.

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TECHNOLOGY FOUNDATION

3. Technologies Used & Rationale

Technology Role in Project Key Features & Implementation Rationale

HTML5 Structure & Layout

Semantic tags (<nav>, <main>, <aside>, <footer>). Accessible modals, clean input forms, and mobile viewport meta configuration.

CSS3 (Glassmorphism) Visual Design System

Modern dark aesthetics with backdrop-filter: blur(16px), glowing borders, CSS variables, CSS Grid, smooth drawer transforms, and toast keyframe animations.

Vanilla JavaScript (ES6+) Application Logic Engine

Zero dependencies or bloated frameworks. Employs async DOM rendering, event delegation, array map/filter/reduce pipelines, and client-side routing.

LocalStorage API Client-Side Database

Browser-level persistent key-value store. Guarantees that carts, orders, and user logins survive page reloads and browser restarts.

FontAwesome 6 Vector Iconography

CDN-loaded SVG icons for shopping bags, star reviews, tech specs, search magnifying glass, lock icons, and
user authentication.

Google Fonts (Outfit) Typography

High-clarity geometric sans-serif typeface tailored for high-end digital tech products and e-commerce
readability.

Git & GitHub Version Control & Cloud Full Git version history, remote tracking, and instant zero-cost static hosting via GitHub Pages.

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FEATURE SHOWCASE

4. Product Catalog & Live Discovery

1. Multi-Category Filtering

• Four dedicated categories: Audio & Sound, Smart Wearables, Computing & Tech, and Accessories.
• Interactive filter chips highlight active states dynamically.
• 'All' button instantly clears filters to show the full collection.
• Real-time array filtering runs in sub-millisecond execution time.

2. Instant Keyword Search

• Live onkeyup search bar with instant query matching.
• Matches keywords across product titles AND detailed descriptions.
• Integrated clear search button ('✕') for rapid reset.
• Displays friendly empty state when no products match.

3. Rich Product Cards

• High-resolution product imagery with hover zoom scale effect.
• Dynamic discount badge (e.g. -20%) auto-calculated from original vs current price.
• Star ratings with real review count displays.
• Price display rendered in Indian Rupees (■) with locale commas.

4. Quick-View Modal Window

• Eye icon trigger opens modal overlay with blurred backdrop.
• Displays full expanded tech specifications and high-res preview.
• Direct 'Add to Cart Now' action from inside modal.
• Responsive 2-column layout that shifts to vertical stack on mobile.

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FEATURE SHOWCASE

5. Interactive Shopping Cart & Price Engine

Slide-Out Cart Drawer Architecture

• Fixed overlay drawer that glides smoothly from the right edge with cubic-bezier
transition.
• Real-time badge counter on navbar bag icon indicates total quantity of items.
• Interactive quantity stepper controls (+ / -) per cart item.
• Automatically purges item from cart when quantity is decremented to 0.
• Empty state banner with clear call-to-action when cart has no items.

LocalStorage State Sync

• Every add, update, or delete operation immediately invokes saveCart().
• Cart contents persist across browser tabs, page refreshes, and computer reboots.

Indian Rupee (■) Pricing Engine

• All prices calculated using standard Indian number formatting via custom formatINR() utility:
return '■' + Number(amt).toLocaleString('en-IN', { ... });
• Automatically formats large figures with proper Indian comma separators (e.g. ■2,999.00).
• Calculates dynamic subtotal and final checkout total.
• Shows free promotional shipping badge for internship demonstration.

Sample Catalog Pricing:

• AeroPulse ANC Headphones: ■1,999.00 (Was ■2,499)
• Titanium Apex Pro Smartwatch: ■2,999.00 (Was ■3,499)
• Mechanical CyberDeck Keyboard: ■1,299.00 (Was ■1,599)
• SonicBass Mini Speaker: ■599.00 | Mouse: ■899.00

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FEATURE SHOWCASE

6. Checkout Process & Order Tracking

1. Multi-Field Shipping Form

• Full Name, Email, Shipping Address, City, and Postal Code.
• Client-side HTML5 input validation prevents blank submissions.
• Auto-populates logged-in user's name and email automatically.
• Order summary breakdown displays total amount before placing order.

2. Unique Order ID Generation

• Generates a cryptographically random 6-digit bill reference number: const orderId = Math.floor(100000 + Math.random() * 900000);
• Records exact ISO timestamp (date & time) for verification.
• Stores complete snapshot of purchased items and prices.

3. Order History Dashboard ('My Orders')

• Dedicated modal accessible via navbar button.
• Lists historical purchase receipts in reverse-chronological order.
• Displays live status badge: Processing.
• Itemized breakdown with item counts, thumbnails, and total bill.

4. Instant User Feedback (Toast Notifications)

• Non-blocking animated toast alerts slide in from the bottom right.
• Success notifications on Add to Cart, Order Placed, Login, and Logout.
• Auto-dismissing after 3 seconds with smooth fade-out animation.

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FEATURE SHOWCASE

7. User Authentication & Session Handling

User Registration (Sign Up) User Authentication (Sign In) Session Persistence

• User supplies username, email, and password (min 6
chars).
• Verifies username uniqueness against nexus_users.
• Registers new user and signs in immediately.
• Displays personalized toast alert.

• Case-insensitive username lookup with password
verification.
• Error toast on invalid credentials.
• Switches dynamically between Sign In and Sign Up tabs in modal.
• Updates Navbar to show user's name.

• Active session stored in nexus_current_user.
• Navbar reveals 'My Orders' button when authenticated.
• Clicking user name prompts quick logout confirmation.
• Guest checkout remains fully supported!

Dual-Mode Accessibility (Guest vs Logged-In User):

• Guest Users: Can immediately browse products, filter categories, search, add items to cart, and checkout without being forced to create an account.
• Authenticated Users: Get automated pre-filling of shipping forms and direct access to order history linked to their specific username.
• Demonstrates modern user-retention design where checkout barriers are minimized.

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ENGINEERING EXCELLENCE

8. Mobile-First & Responsive Optimizations

Two-Row Mobile Navbar

• On screens < 768px, Navbar splits cleanly into two rows:
- Row 1: Logo on left, User & Cart buttons on right.
- Row 2: Full-width search bar with magnifying glass icon.
• Follows modern standards used by Amazon, Apple, and Flipkart.

Touch-Optimized Category Bar

• Implements -webkit-overflow-scrolling: touch.
• Hidden scrollbars with horizontal kinetic swipe gestures.
• Touch manipulation tags prevent double-tap delays on buttons.

Full-Screen Drawer & Bottom-Sheet Modals

• Cart drawer slides to 100% viewport width on phones.
• Modals adapt to elegant bottom-sheets with rounded top corners.
• Form inputs locked at font-size: 16px to prevent unwanted iOS Safari auto-zoom behavior.
• Single-column stacked product cards with large touch targets.

Mobile Browser Header Theme Integration

• Configured <meta name='theme-color' content='#090d16'>.
• Mobile address bars in Chrome Android and iOS Safari blend natively with the app's dark
glass background.

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PROBLEM SOLVING

9. Key Challenges Faced & Engineering Solutions

Engineering Challenge Root Cause / Complication Adopted Technical Solution

Serverless Data Persistence

Removing the Python/Django backend eliminated SQL tables for storing carts and orders.

Designed a normalized JSON schema layer inside HTML5 localStorage with auto-initialization and storage versioning.

Stale LocalStorage Caches

Users who loaded older builds still had obsolete dollar prices cached in browser memory.

Implemented STORAGE_VERSION = 'nexus_v2_inr' check in initStorage() to automatically refresh cached catalogs.

Mobile Form Zoom Bug in iOS

Tapping input boxes on iPhones caused Safari to
automatically zoom the page, distorting UI.

Enforced strict font-size: 16px !important on all input, select, and textarea elements under @media (max-width: 768px).

Horizontal Viewport Overflow

Fixed-width 400px cart drawer caused horizontal scrollbars on smaller 360px mobile viewports.

Applied width: 100%; max-width: 100vw; right: -100% clamped styles on mobile breakpoints.

Indian Rupee Formatting

Default JavaScript formatting uses Western millions/billions commas instead of Indian lakhs.

Created custom formatINR() using toLocaleString('en-IN') for authentic
Indian Rupee currency representations.

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CODEBASE & DELIVERY

10. Repository Structure & Deployment

Minimalist Zero-Dependency Repository:

CodeAlpha_ECommerceStore/
■■■ index.html # Semantic Single-Page Application
■■■ styles.css # Full Glassmorphic Design System
■■■ app.js # Client State, Logic & DB Engine
■■■ README.md # Complete Documentation & Guide

Why this structure is optimal:

• Zero installation overhead — runs instantaneously on double click.
• Clean separation of structure, aesthetics, and business logic.
• Extremely lightweight bundle size (~50 KB total) guarantees near-instant load speeds.

GitHub Repository & Deployment:

Official GitHub Repo:
https://github.com/shauryakartik2-source/Ecommerce-store

GitHub Pages Deployment Workflow:

1. Repository pushed to GitHub main branch.
2. In GitHub Settings ➔ Pages ➔ Source: 'Deploy from a branch'.
3. Branch set to main and folder to / (root).
4. Within 30 seconds, the site is live on global CDN with free SSL certificate!

Verification & Git Status:

All commits verified with clean working tree, untracked pycache purged, and synced with origin/main.

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FINAL ASSESSMENT

11. Conclusion, Learnings & Future Roadmap

Key Internship Learnings:

• Deep DOM & Event Handling: Mastered vanilla event delegation, dynamic
template rendering, and modal controllers.
• Modern CSS Architecture: Built an end-to-end design system using
glassmorphic backdrops, CSS variables, and fluid responsive grids.
• Client-Side State Modeling: Learned how to design robust, persistent multi-model state architectures using the Web Storage API.
• Mobile User Experience: Designed adaptive layouts that gracefully transition between phone, tablet, and widescreen layouts.

Personal Growth:

Developed self-reliance in building production-quality web applications from ground up with clean, maintainable code standards.

Future Roadmap & Enhancements:

• Payment Gateway Integration: Connect Razorpay or Stripe API for real online UPI, Card, and Netbanking transactions.
• Progressive Web App (PWA): Add service workers and manifest for offline browsing and home-screen app installation.
• Backend Microservice (Optional): Connect Python Django REST or Node.js server for cloud sync and admin order fulfillment.
• User Reviews & Ratings: Allow customer feedback and product rating submission directly in the browser.

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THANK YOU! Shaurya Kartik | B.Tech CSE | Roorkee Institute of Technology | CodeAlpha Intern Questions & Discussions are warmly welcomed.

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